

Public Economics and Development

Week 8: Evolution of tax structures in developing countries

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April 17th, 2026

Challenges in tax administration: Information

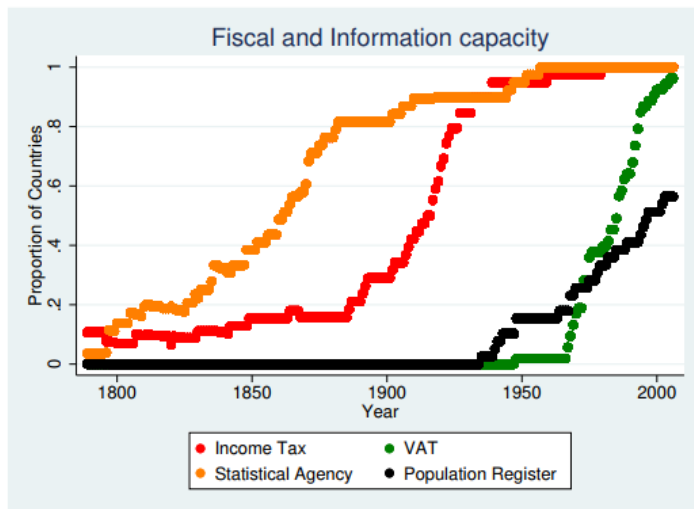
Information on taxable transactions are crucial for imposing tax

- Having a firm registered
- Income sources with traceable sources
- Formal transactions with verifiable paper trails

Modern economies implement technologies and systems gathering these information

- Statistical offices & population registers compile info on various economic activities
- Employee-employer relationships enable accurate tracking of income
 - ▶ Third-party information (not self-declared information) used as source
- Value-added tax (VAT) creates a paper trail between different levels of supply chain
 - ▶ VAT charged on sales - input for each firms
 - ▶ Downstream firms have incentive to ask their suppliers for a receipt
 - ▶ Since the receipt provides evidence for input costs, which can lower their tax burden

Rise of informational capacity and taxation over time



Fiscal and Information Capacity in a sample of 39 countries

Lack of tax administration capacity in developing countries

Limited access to necessary information

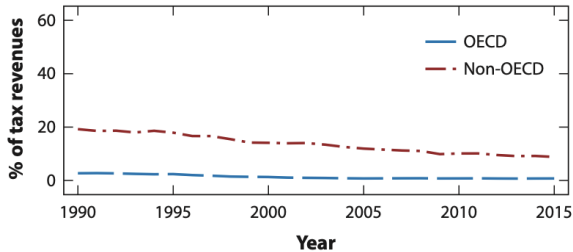
- Large share of economic activities are taking place in informal markets (last week)
- Usage of third party information is limited
 - ▶ Higher share of self-employed individuals (who self-declare income)
 - ▶ Less transactions based on digital tools such as credit cards (cash do not leave records)
- Tax agents are often under-trained or under-monitored

What differences does it lead to

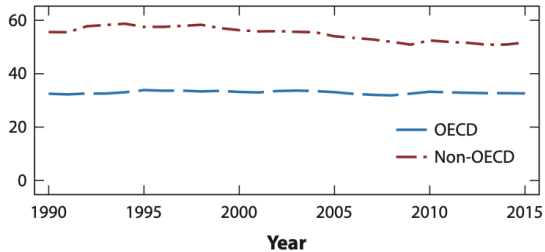
- Generally lower share of taxes relative to economy
- High share of trade taxes such as tariffs (easier to monitor)
- Collect more taxes from firms than individuals (fewer firms than individuals)

Taxes in OECD vs non-OECD countries

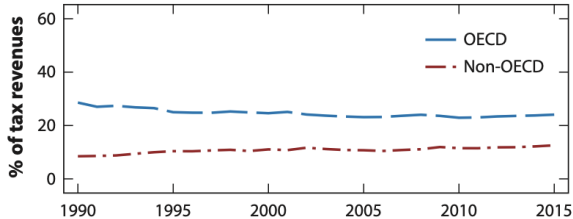
Taxes on imports and exports



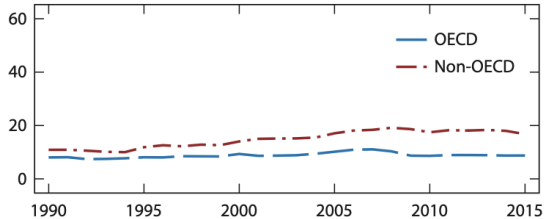
Taxes on goods and services



Personal income tax



Corporate income tax



New developments in tax technology

What is being tried in developing countries

- Expansion of digital money and banking system for transparent transactions
- Increases in digitalized identification of properties and individuals
- E-filing: Making compliance to taxation easier

Some new studies:

- Effects of the assignment of tax collectors
- Effects of incorporating local knowledge in taxation
- Implementation of various tax campaigns to increase compliance from taxpayers

Roadmap for today

1. Overview setup of the topic

2. Evolution of economic structure and taxation

- 2.1 Oyebola, Tourek (2024): How Can Lower-Income Countries Collect More Taxes? The Role of Technology, Tax Agents and Politics
- 2.2 Jensen (2022): Employment Structure and the Rise of the Modern Tax System

3. How tax administration affects tax collection: Evidence from experiments

- 3.1 Pomeranz (2015): No Taxation without Information: Deterrence and Self-Enforcement in the Value Added Tax
- 3.2 De Neve, Imbert, Spinnewijn, Tsankova, Lutz (2021): How to Improve Tax Compliance? Evidence from Population-Wide Experiments in Belgium

4. Effect of tax collectors on compliance

- 4.1 Basri, Felix, Hanna, Olken (2021): Tax Administration versus Tax Rates: Evidence from Corporate Taxation in Indonesia
- 4.2 Balán, Bergeron, Tourek, Weigel (2022): Local Elites as State Capacity: How City Chiefs Use Local Information to Increase Tax Compliance in the Democratic Republic of Congo

Oyebola, Tourek (2024):
How Can Lower-Income Countries Collect
More Taxes? The Role of Technology, Tax
Agents and Politics

Q: How do you improve tax system in developing countries?

Increasing tax revenues is a major goal in many low/middle income countries

- Generally have lower share of their GDP as tax revenues
- Low taxation is a symptom of economic and institutional constraints
 - ▶ Tax base not large enough, lack of constraint on redistribution/taxes
- Does economic growth automatically expand tax capacity? Not exactly
 - ▶ Separate investments in improving the tax system required

This paper reviews recent developments in tax administration in developing countries

- Identifying taxpayers (expanding tax registries and databases)
- Detecting tax burden (usage of third-party information)
- Collect taxes to state coffers (Billing systems, legal measures for noncompliance)

Technology in tax collection

Identification: Cost of linking identity info across government agencies and registries↓

- National ID (Ghana Card), Digital mapping of properties (Liberia)
- ID information may need to be combined with other measures (Okunogbe 2023)

Detection: Collect third-party info (declaration of tax liabilities by third person)

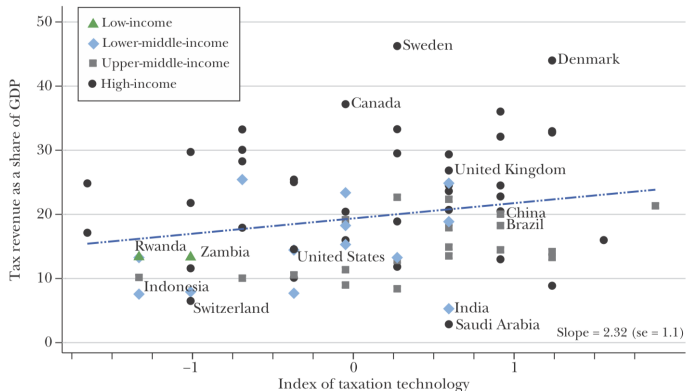
- 'Digital trails' from value-added taxes and employment taxes
- Requires countries to cross-check information using multiple sources
- Need to close international loopholes

Collection: Increases in electronic filing and mobile payment options

- Reduces cost of monitoring and Compliance
- May disadvantage those less familiar with electronics
(rural, less-educated, less-established firms)

Cross-country evidence on technology and collection

Taxation Technology and Tax-to-GDP



Source: Tax revenues as a share of GDP measure drawn from the International Survey on Revenue Administration (CIAT, IMF, IOTA, and OECD 2022), using values for 2018. Taxation technology measure drawn from OECD (2023) Global Survey on Digitalisation.

Role of tax collectors

What determines the effectiveness of officials

- Deployment: low citizen-to-staff ratio and appropriate allocation
 - ▶ Low burden per staff
 - ▶ Specialization: Matching staff depending on tax types and taxpayer traits
- Incentives: Tie staff revenues to tax revenue generations
 - ▶ Risk: Could increase scope for corruption
 - ▶ Monitoring staff with audits crucial

Interaction with technology

- Substitute? Obviate need for in-person Interaction
- Complement? Help expedite tedious processes and collecting contextual knowledge
- Balancing the two forces is key (and unresolved question)
- Resistance to adopting the technological change

Political incentives for taxation

How do political incentives determine tax collection

- Tax $\uparrow$ $\rightarrow$ fund public services, social programs, infrastructure $\rightarrow$ support re-election
- Tax $\uparrow$ $\rightarrow$ Citizen demand, expectation for accountability $\uparrow$ $\rightarrow$ not help that much

What are the factors shaping success of tax collection?

- Alternative revenue: Dampen motivation for Taxation
- Political competition: Competition for better governance (through public services)
- Availability of tech: Reduce cost of collection and resistance from taxpayers

Expanding our knowledge

Findings from other relevant studies

- Tax revenue and compliance increases with increased tech and enforcement
- Many results are from small-scale local experiments: Generalizable?

What remains?

- How to make technology and personnel complementary?
- How to tax new commodities (digital activities, digital currencies)?
- How to increase political support for taxation?

Jensen (2022):
Employment Structure and the Rise of the
Modern Tax System

Q: How do modern tax systems rise?

Modern tax systems are characterized by increased direct taxes (income tax)

- Requires high collection capacities
- Changes in the structure of employment may explain increase in collection
 - ▶ Enforcement in employer-employee structure > Self-employed
 - ▶ Existence of third-party information (employer, in case of income tax)

This paper uncovers new facts on tax capacity with microdata across countries and time

- Employee share increases over income distribution & further down the income distribution as country develops
- Exemption threshold gradually decreases in income distribution over development
- Threshold is located such that employee share on tax base remains constant and high

Data, definition, and calculation strategy

Data source and definitions

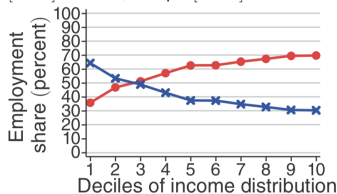
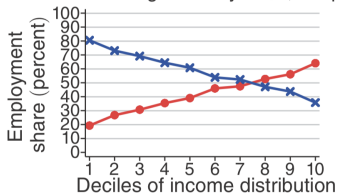
- Household surveys (100 countries)
 - ▶ Employment type of economically active population (EAP)
 - ▶ Information on taxable earnings (employment / self-employment / capital / other income)
 - ▶ For US (1870-2010) and Mexico (1960-2010), compile long run time series

Definition of key terms and strategies for calculation

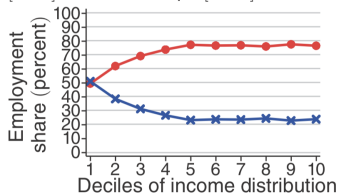
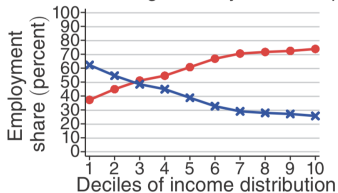
- Employee: if work type generates third-party information usable in tax enforcement (Otherwise, self-employment)
 - ▶ Self-employed: casual laborers, own account workers, workers w/o contracts
 - ▶ Calculate share of employees in EAP across country's income distribution (in deciles)
- Threshold: Minimum level of gross income above which individuals have to pay taxes
 - ▶ From country tax reports of International Bureau of Fiscal Documentation
- Income tax base: % EAP whose gross income is above income tax exemption threshold

Fact1: Employee share $\uparrow$ with income and development levels

Profile for average country at \$3,239 pc [LHS] and \$5,796 pc [RHS]



Profile for average country at \$8,826 pc [LHS] and \$11,257 pc [RHS]

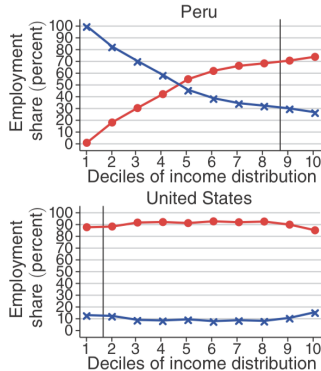
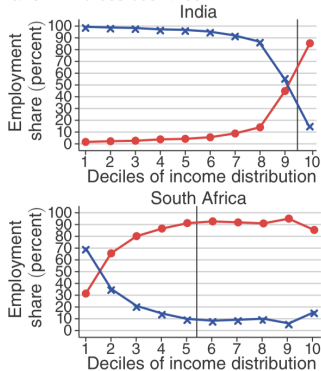


Red: Employee, Blue: Self-employed

- Within country, employee share increase over income
- Across countries, employee share profile moves leftward
- Over transition, increase in employee share concentrated in deciles gradually further down

Fact2: Exemption threshold moves downward as country develops

Panel A. Across countries



Red: Employee, Blue: Self-employed, Vertical: Threshold

- At low development, threshold at top deciles
- With development, growth in employee share comoves with falling thresholds
- Tax base increases
- Documents frequent updating of thresholds (base increase not driven by nominal inflation)

Fact3: Threshold is located where employee share remains constant

Panel C. Constant employee composition on income tax base

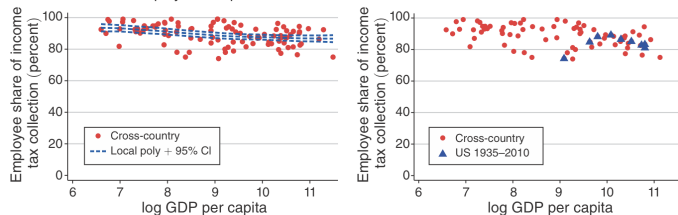


FIGURE 4. SIZE AND COMPOSITION OF INCOME TAX BASE OVER DEVELOPMENT

- Composition of tax base is constant and high in employee share
- Employee share: 85-95%
- Suggest that employee composition matters in deciding threshold
- % Self-employed $\uparrow \rightarrow$ Enforcement costs $\uparrow$ (high employee share desirable)

Takeaways from the stylized facts

Cost channels at play in expanding tax base

- Concerns with taxing income segment with too high-self employed population
 - ▶ Employees have information trails, self-employed do not
- Compliance cost imposed on newly taxed self-employed may be high
 - ▶ Employees have third-party withholding
- Worsen between-group horizontal equity among employee vs self-employed
 - ▶ Self-employed have evasion opportunities → effective taxes paid < legal liabilities

Rise of modern taxation: Decrease in threshold (broad-based tax) happens when

- Increase in employee share at lower end of income distribution (Fact 1)
- This lowers costs of reform and expand tax base (Fact 2)
- While retaining high employee share, making tax administration manageable (Fact 3)

Implications for developing countries

Redistribution by direct taxes might be ill-suited for developing countries

- Developed countries extensively use third-party information
 - ▶ Evidenced by increase in employee share down the income distribution over time
- Developing countries cannot enforce taxes flexibly due to cost constraints
 - ▶ Higher share of self-employment (difficult to gather info)
- For such settings, indirect taxation could be the best option for redistribution

Remaining issues

- How would implementation of technology affect tax base long-run?
 - ▶ More use of third-party information → Rising 'employee' share?
 - ▶ Expanding tax base would then be less costly → tax collection, public goods ↑
 - ▶ But what about adoption (existence of political will)?

Pomeranz (2015):

No Taxation without Information: Deterrence
and Self-Enforcement in the Value Added Tax

Q: How does third-party information in VAT help taxation?

VAT is believed to facilitate enforcement via a third-party information

- Built-in incentives to ensure that transactions are revealed
 - ▶ Downstream firms demand receipt in order to deduct input costs from their liabilities
 - ▶ This incentive is sustained until the transaction reaches the final consumer
- Implementation: 47 countries in 1990 → 140 in 2010s
- But the effect of this built-in structure not empirically tested at this point

This paper uses randomized experiments to verify the properties of this structure

- This experiment was conducted on more than 400,000 firms in Chile
- Letter message experiment to test preventive deterrence effect on evasion
- Spillover experiment to identify underlying mechanisms behind information effects
- Idea is to test how the paper trails interact with tax audit probabilities

Conceptual framework of VAT

Setup

- Let $i \in \{1, 2, \dots, N\}$ index the firm in a value chain where firm $i + 1$ buys from firm i
- s_i and c_i are the true sales and costs for firm i . $\hat{s}_i$ and $\hat{c}_i$ are reported sales and costs
- Firm $i + 1$ wants receipt from firm i to deduct s_i from its cost c_{i+1} (No collusion here)
- The last firm N does not face this problem: Consumers do not require receipt

Types of potential evasion and role of perceived audit probabilities

- Unilateral: Firms underreport own liability while risking discrepancy with other firms
 - ▶ $\hat{c}_i > c_i$ or $\hat{s}_i < s_i \implies \hat{c}_i > \hat{s}_{i-1}$ or $\hat{s}_i < \hat{c}_{i+1}$
- Collusive: Both firms drop the sale entirely or underreport liabilities
 - ▶ $\hat{c}_i \neq c_i$ but $\hat{c}_i = \hat{s}_{i-1}$ or $\hat{s}_i \neq s_i$ but $\hat{s}_i = \hat{c}_{i+1}$
 - ▶ If firm n engage in this evasion, it must be the case for all $i \geq n$ up to $i = N$
 - ▶ It must also be that $\hat{s}_i - \hat{c}_i \leq s_i - c_i \iff c_i - \hat{c}_i \leq s_i - \hat{s}_i$
- As audits can uncover paper trail, the threat of audit should deter evasion

Prediction on tax evasion behaviors after announcement of audit

Unilateral evasion

- Suppose audit is announced on firm n in the middle of the chain
- This forces supplier and client firms to revise perceived audit probability
- Thus, they revise declared transactions to be consistent with firm n
 - ▶ Supplier: Increase sales figures and VAT payment
 - ▶ Client: Decrease cost figures and increase VAT payment

Collusive evasion

- Again, firm n receives audit announcement
- Firm n would be incentivized to create a paper trail (increased sale and costs)
- Supplier firm induced to increase reported sales and VAT payment
 - ▶ Possible chain reaction continues to onto its own supplier
- Client firms has room to increase their cost and reduce VAT
 - ▶ Since supplier raised sale, client has more room to declare its costs

Experiment design

Experiment 1: Letter experiment to check behavior of a single firm

- Deterrence: Reminds firms of the possibility of audit (102,031)
- Tax Morale: Highlight social norms on taxes, but no content on deterrence (18,579)
- Placebo: New feature on tax website to see if messaging alone has impacts (18,519)
- Outcome variable: VAT filings of each firms

Experiment 2: Spillover experiment to see if tax compliance on other firms are affected

- Sample of 5,600 suspected firms already chosen for audit
- Randomly pre-announce audit to half of the firms
- Data on partner firms collected in the audit
- Check VAT declarations before and after announcement

Deterrence message increases compliance (reduce evasion)

	Mean VAT (1)	Median VAT (2)	Percent VAT > previous year (3)	Percent VAT > predicted (4)	Percent VAT > zero (5)
Deterrence letter $\times$ post	-1,114 (2,804)	1,326*** (316)	1.40*** (0.12)	1.42*** (0.10)	0.53*** (0.09)
Tax morale letter $\times$ post	-1,840 (6,082)	262 (666)	0.40 (0.25)	0.30 (0.22)	0.44** (0.20)
Placebo letter $\times$ post	835 (6,243)	383 (687)	-0.11 (0.26)	-0.19 (0.23)	-0.14 (0.20)
Constant	268,810*** (1,799)	17,518*** (112)	47.50*** (0.07)	48.27*** (0.07)	67.30*** (0.06)
Month fixed effects	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	No	Yes	Yes	Yes
Treatment assignment	No	Yes	No	No	No
Observations	7,892,076	1,221,828	7,892,076	7,892,076	7,892,076
Number of firms	445,734	445,734	445,734	445,734	445,734
Adjusted R^2	0.40		0.14	0.28	0.47

Preannouncement also affects partner firms' taxation response

	Percent VAT > previous year (1)	Percent VAT > predicted (2)	Percent VAT > previous year (3)	Percent VAT > predicted (4)	Percent VAT > previous year (5)	Percent VAT > predicted (6)
Audit announcement × post	2.41** (1.14)	2.03* (1.11)				
Audit announcement × supplier × post			4.28*** (1.54)	3.92*** (1.50)	4.14*** (1.52)	3.83*** (1.52)
Audit announcement × client × post			-0.26 (1.64)	-0.28 (1.51)	-0.14 (1.67)	-0.28 (1.55)
Supplier × post			-0.64 (1.62)	0.34 (1.59)	-1.11 (1.67)	0.60 (1.64)
Constant	52.07*** (0.95)	49.06*** (0.94)	52.07*** (0.95)	49.06*** (0.94)	52.75*** (0.96)	50.11*** (0.96)
Controls × post	No	No	No	No	Yes	Yes
Controls × audit announcement × post	No	No	No	No	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	45,264	45,264	45,264	45,264	44,288	44,288
Number of firms	2,829	2,829	2,829	2,829	2,768	2,768
Adjusted R^2	0.05	0.11	0.05	0.11	0.05	0.10

Key takeaways and discussion

Key findings

- Paper trail in VAT self-enforces accurate tax information
- Transactions already under paper trail more likely to comply with audit possibility
- Increasing audit probability increases tax payment of supplier firms
 - ▶ Not necessarily downstream firms: Consumers have less incentives to ask for receipt

Overall message

- Interaction of information and enforcement matters
- Information from paper trail can be a powerful tool
 - ▶ Either in form of VAT or digitalized property registers or earnings
- But it also explains why tax collection in developing countries are low
 - ▶ Many home-based activities with no paper trail
 - ▶ Lack of credible threat/perception of enforcement

De Neve, Imbert, Spinnewijn, Tsankova, Lutz
(2021):

How to Improve Tax Compliance? Evidence
from Population-Wide Experiments in Belgium

Q: How do different administrative approaches affect tax compliance?

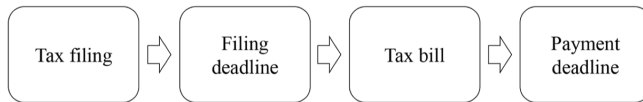
Tax compliance matters for healthy functioning societies

- Cost of noncompliance in US: 319 billion USD annually
- Truthfulness and timely payment both matters for compliance
- Increasing yet incomplete understanding on nonpecuniary motives for tax compliance
 - ▶ Recent interest in 'tax morale'

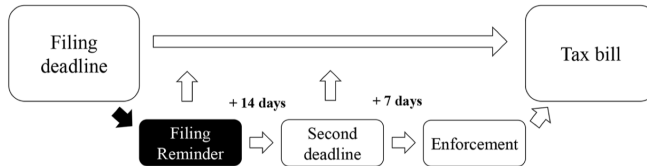
This paper studies how communications from tax authorities affects tax compliance

- Different dimensions: Simplified communication vs deterrence vs appeal to tax morale
- Messaging experiment from tax authorities in Belgium (2014-16)
- Estimates how different dimensions of tax reminders differentially affect compliance
 - ▶ Message content
 - ▶ All taxpayers vs taxpayers who missed deadline

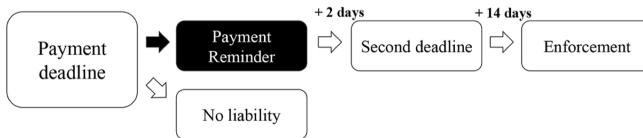
Tax process in Belgium



(a) Filing and payment



(b) Filing reminder process



(c) Payment reminder process

Experiment design

Treatment

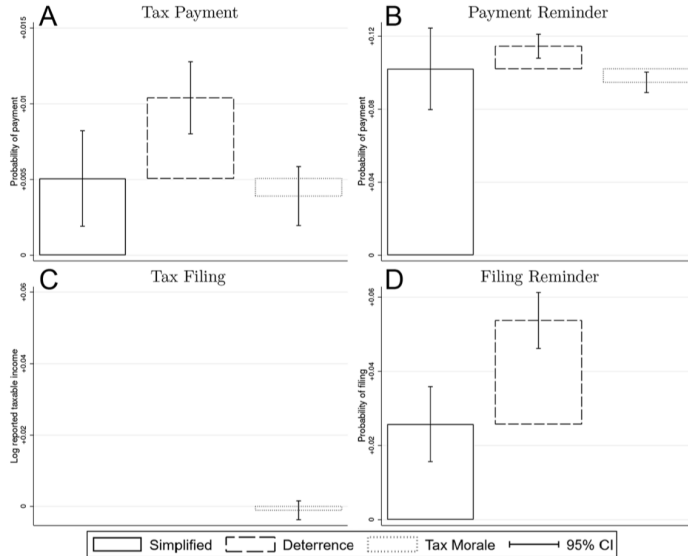
- Simplification: Shorter letters, reduce information overload, action-relevant info
- Deterrence: Simplified + make penalties explicit
- Tax deterrence: Simplified + emphasize social benefits of taxation
- Control group: Standard long message with complicated information

Implementation of the experiment

- Four experiments: Filing, Filing reminder, Payment, Payment reminder
 - ▶ Filing experiment only had tax morale
 - ▶ Filing reminder experiment did not have morale component
- Only the ones who missed the deadlines are part of the reminder experiment
- Randomized based on birthday (payment) and national ID number (other experiments)
- Regression is an OLS where

$$y_i = \alpha + \beta_s S_i + \sum_j \beta_j T_i^j + \gamma X_i + e_i$$

Different effects on tax payment



Exploring mechanisms

Simplification works primarily with extensive margins

- Make relevant deadline salient: Effects larger around the deadlines
- Does not lead to large increases in the amount of taxes paid

Deterrence component complements the effect of simplified messages

- Makes deadline and relevant information salient and encourage immediate action
- No large effect on the amount of taxes paid
- Tax morale part may have increased relevant knowledge but ineffective beyond this

Heterogeneity

- Simplification more effective on households with children and median solvency score
- Deterrence most effective on young taxpayers, and those with low liabilities

Cost-effectiveness of interventions and remaining discussion

Cost effectiveness of different interventions

- The authors estimate the costs using RD estimates around enforcement intensity
- Based on this cost-effectiveness is measured in three ways
- In all of these results: Costs of simplification < Costs of extra enforcement
 - ▶ Simplification: Total cost is 10% of extra deterrence measure in one of the approaches

Discussions

- Compliance can increase if directions are straightforward
- Morale message not effective: Likely because people are aware of importance of taxes
- Lack of compliance, thus, is largely explained by inattention of taxpayers
- Can this apply on tax collectors?
 - ▶ They know that as public servants, collecting taxes are crucial
 - ▶ But they may be technically unaware of deviant behaviors...

Basri, Felix, Hanna, Olken (2021):
Tax Administration versus Tax Rates: Evidence
from Corporate Taxation in Indonesia

Q: How much does improving tax administration raise tax revenues?

Tax revenues in developing countries remain low

- Is it due to high elasticity of taxable income (ETI)?
 - ▶ High ETI implies that tax revenues fall in large amounts w.r.t. rise in tax rates
- Or is it a problem of tax administration?
 - ▶ Enhanced administration may make evasion/avoidance much more difficult
 - ▶ Rise in tax revenues without rise in tax rates

This paper studies this question in context of corporate taxes in Indonesia

- Administrative reforms created medium size taxpayer offices (MTO)
 - ▶ higher ($\times 3$) staff-to-taxpayer ratio than other offices to increase enforcement
- First study the effect of intensified administration in administrative tax data
- Then compare with changes in statutory corporate income tax rates several yrs later

Relevant tax reforms in Indonesia

Administrative reforms in Indonesia in mid-2000s

- Moved several hundred top corporate taxpayers across 19 tax regions to MTOs
- MTOs exclusively focused on large taxpayers and had more staff than other offices
- Taxpayers not assigned to MTOs assigned to smaller offices (PTOs)
- Staffs in MTOs and PTOs are similar in experience and performance assessments

Tax rate reform on corporate income tax

- In 2009, Indonesia changed from progressive 15-30% rate to flat 28% tax rate
- Pre-reform tax base was net taxable profits
- Post-reform taxes are determined based on gross revenues
- Total tax and discount determined by a set formula across 3 revenue brackets

Data sources and regression

Data sources

- Anonymized microdata on corporate taxpayers (2003-2011)
- Detail on tax reporting, wage bills, audits, VAT assessments, and tax payment

Regression approach

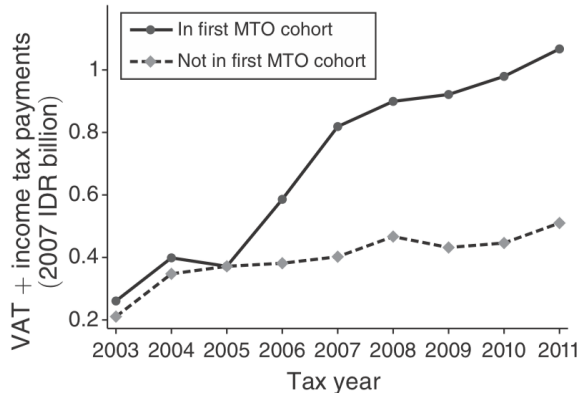
- MTO reform: Use matched DID

$$y_{it} = \alpha + \beta(M_i \times 1[t > 2005]) + \delta_t + \delta_i + e_{it}$$

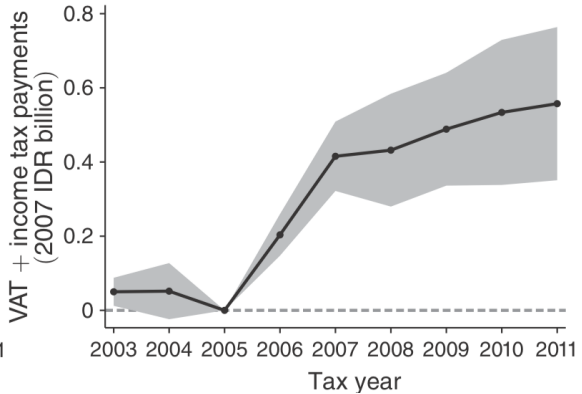
- ▶ Control: Taxpayers in same region with similar pre-MTO tax payments and gross revenue
- ▶ Match control and treatment based on entropy balancing (Hainmueller 2012)
- Effects of tax rates: Instrument change in tax rates with tax formula following reform
 - ▶ The goal is to estimate elasticity of taxable income w.r.t tax rate changes
 - ▶ Lower elasticity implies that taxable income does not respond much to tax rate
 - ▶ Used to calculate how much tax rate should change to match revenue from MTOs

Tax revenues following MTO nearly doubles

Panel A. MTO versus non-MTO weighted annual averages

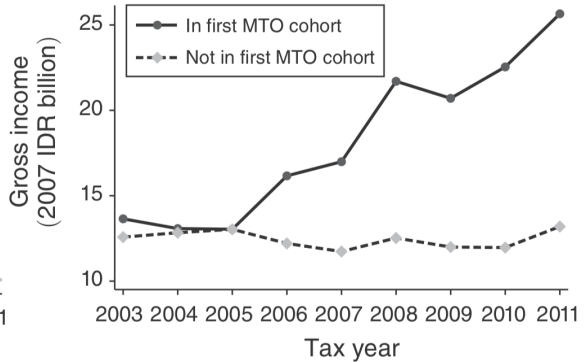
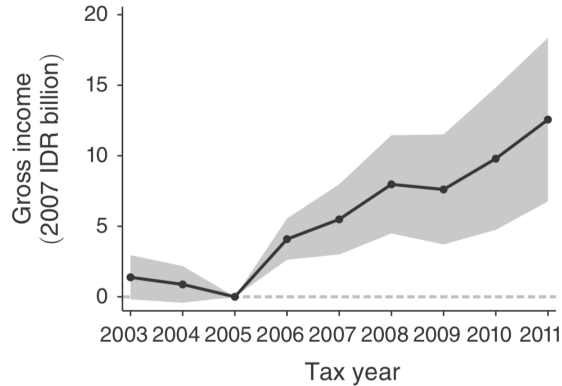


Panel B. Year-by-year estimates



Effects driven by higher reported incomes

Panel A. Gross income



ETIs are actually similar across different firms

TABLE 6—ESTIMATED ETI WITH RESPECT TO THE
NET OF TAX RATE

	Instrument: reform-induced change in MTR		
	Separate by MTO status		
	All taxpayers (1)	MTO (2)	Not MTO (3)
<i>Panel A. First stage</i>			
Endogenous: $\Delta \ln(\text{net of tax rate})$	0.979 (0.010)	0.981 (0.018)	0.981 (0.010)
<i>F</i> -statistic	10173.650	3000.712	9075.552
Observations	12,816	726	12,090
<i>Panel B. IV (ETI estimates)</i>			
Outcome: $\Delta \ln(\text{taxable income})$	0.579 (0.198)	0.344 (0.380)	0.764 (0.214)
<i>p</i> -value of difference	0.335		
Taxpayer fixed effects	Yes	No	No
Year fixed effects	Yes	Yes	Yes

- ETI is 0.579, similar to many developed countries
- ETIs statistically similar = MTOs did not work by reducing ETIs

Other key findings and discussion

Other key findings

- Tax enforcement no longer size-dependent
 - ▶ Before: Small offices focusing on large firms
 - ▶ This created disincentives for small firms to grow
 - ▶ Now: MTOs able to treat firms of various sizes equally
- Tax revenue increases by administrative reform can be cost-effective
 - ▶ Improved tax administration = raising top rates on all firms by 8 percentage points
 - ▶ So improving administration can have significant returns for developing countries

Discussion: How to improve tax administration

- Improving staff-to-taxpayer ratio by raising staffing
- Alternatively, specialized offices may help
- Another dimension is the quality of staffing (education, capital, training)
 - ▶ Better trained, well-equipped staffing can improve effective number of personnel

Balán, Bergeron, Tourek, Weigel (2022):
Local Elites as State Capacity: How City Chiefs
Use Local Information to Increase Tax
Compliance in the Democratic Republic of
Congo

Q: How can local information be used effectively to increase taxes?

How can fiscal capacities develop in presence of local & traditional elites

- On the one hand, can capture politics and civic society
- On the other, could be room to collaborate to improve governance
- Potential trade-offs
 - ▶ Enforcement backed by the state vs local knowledge and coordination?

This study investigates if local elites can contribute to statebuilding

- Comparison between state agents vs local elites
 - ▶ Local elites know more contextual knowledge but could lead to leakage in revenues
 - ▶ State agents have better enforcement capacity
- Based on field experiment randomizing type of tax collectors across neighborhoods
- Estimate changes in revenues and bribes, with tests for various mechanism

Taxation in Democratic Republic of Congo (DRC)

Setting of DRC

- Low-capacity state: tax/GDP ranks 188 out of 200 and \$0.30 per capita tax revenue
- Relies on property tax: First campaign to expand property tax in 2016
- Public goods and services are scarce and low-quality
 - ▶ Limited running water, electricity, unrepaired roads etc

The current study focuses on the second campaign (May-Dec 2018)

- Collectors trained by provincial ministry, separate for state and chiefs
- Involved property registration updates, providing liability information
- Followed by tax visits (revisit until taxes are paid)
- Collectors compensated based on houses registered and revenue collection
- Property tax rates are tiered flat, fixed fees (low: \$2 tax, 89% vs high: \$9 tax, 11%)
 - ▶ Equivalent to 0.32% of property value ($\simeq$ lower end of property tax rates in US)

Data and empirical strategy

RCT on 45000+ properties in 356 neighborhoods in Kasai-Central

- Randomized at neighborhood, treatment at past campaigns, experience of chiefs
- Central (state agents) vs Local (chiefs) vs State agents w/ local info vs state and local
- Pure control group: Declare their property values voluntarily
- Other aspects of collection equal
 - ▶ Property registration, assessment, tax liabilities, training, collection compensation
- Complemented with tax administrative data and household surveys

Estimation leverages random assignment of treatment (indiv i , neighbor j , strata k , time t)

$$y_{ijkt} = \beta_0 + \beta \text{Local}_{jkt} + \gamma X_{ijkt} + \alpha_k + \theta_t + e_{ijkt}$$

- For mechanisms, other treatment arms included as separate variable

Increase in compliance and revenues

TABLE 4—LOCAL VERSUS CENTRAL: COMPLIANCE AND REVENUES

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Compliance</i>					
Local	0.023 (0.008)	0.032 (0.007)	0.032 (0.008)	0.033 (0.007)	0.040 (0.008)
Observations	28,872	27,764	213	27,764	23,803
Clusters	221	213		213	213
Central mean	0.068	0.063	0.065	0.063	0.073
<i>Panel B. Revenues</i>					
Local	57.627 (25.688)	79.640 (22.856)	81.830 (38.595)	68.855 (20.560)	81.991 (23.562)
Observations	28,872	27,764	213	27,764	23,803
Clusters	221	213		213	213
Mean	192.891	182.236	210.134	182.236	208.568
Time fixed effects	No	Yes	Yes	Yes	Yes
House fixed effects	No	No	No	Yes	Yes
Stratum fixed effects	Yes	Yes	Yes	Yes	Yes
Exempt excluded	No	No	No	No	Yes

Accurate assessments & trust in government ↑, but bribes ↑

TABLE 5—LOCAL VERSUS CENTRAL: MISMANAGEMENT AND VIEWS OF GOVERNMENT, CHIEFS, AND TAXES

	$\hat{\beta}$	SE	R^2	Observations	$\bar{x}_{Central}$
<i>Panel A. Property assessments</i>					
Assigned exemption	0.039	0.021	0.055	13,772	0.266
Incorrect exemption	−0.012	0.007	0.020	13,771	0.044
Assigned high band	0.030	0.021	0.230	27,764	0.114
Incorrect assignment	−0.013	0.006	0.041	27,764	0.031
<i>Panel B. Bribes</i>					
Paid bribe (midline)	−0.001	0.003	0.007	18,596	0.016
Gap self versus admin (midline)	0.016	0.009	0.018	14,309	0.077
Paid bribe (endline)	0.018	0.009	0.049	1,169	0.014
Other payments (endline)	0.031	0.014	0.041	2,407	0.094
<i>Panel C. View of government</i>					
View of government (index)	0.023	0.049	0.100	2,411	0.011
Trust in government	0.127	0.057	0.075	2,286	0.028
Responsiveness of government	−0.049	0.045	0.099	2,282	0
Performance of government	−0.060	0.052	0.060	2,179	−0.014
Integrity of government	0.043	0.047	0.058	2,313	0.016
<i>Panel D. View of taxation</i>					
Perceived tax compliance on avenue	0.100	0.055	0.073	1,851	0.026
Trust in tax ministry	0.085	0.061	0.073	2,259	0.025
Property tax morale	0.075	0.047	0.057	2,343	0.014
Fairness of property taxation	−0.004	0.053	0.046	2,407	0.003
Perception of enforcement	−0.019	0.058	0.070	2,379	0.015

Evidence of local information at play

TABLE 7—CENTRAL VERSUS CLI

	Compliance (1)	Revenues (2)	Visited (3)	Visits (4)	Compliance (5)	Compliance (6)
CLI	0.024 (0.009)	46.566 (21.200)	−0.016 (0.028)	−0.026 (0.044)	0.026 (0.014)	0.022 (0.009)
Local						0.046 (0.007)
Visit control	No	No	No	No	Yes	No
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
House fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Stratum fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,636	20,636	13,884	13,877	5,283	33,746
Clusters	165	165	163	163	161	267
Central mean	0.051	150.66	0.387	0.497	0.097	0.052
Test CLI = Local (<i>p</i> -value)						0.007

Discussion and takeaways

What explains the chiefs' advantage? Better targeting with information advantage

- Lower effort cost? Chiefs did not make more visits
- Target their visits? When state agents received info from chiefs, compliance $\uparrow\uparrow$
- Better persuasion? Neutralize info advantage (visit in linear route) $\Rightarrow$ Central = Local

Takeaways and remaining questions

- Should countries with low capacities rely on local elites?
 - ▶ Raise revenues, but also bribes
 - ▶ To shift to state, value of \$1 lost in bribes $\gg$ \$1 gained in revenues
- For technology to increase taxes, need to surpass some state capacity threshold?
 - ▶ Speak to the complementary/substitutional relationship between tech and person

Unanswered questions

Effects of technological advancements on taxation

- More countries adopting digital registries, automated paper trails, and identification
- Many evidence that it led to better detection and compliance
- Other questions that remain not yet fully explored
 - ▶ Can it lead to better training of tax officials?
 - ▶ Can it lead to change in perception about taxation among citizens?

What can better administration of taxes do

- More efforts to increase administration in place
- But what about administration for how to *use* these taxes?
 - ▶ How can developing countries provide public services efficiently with these taxes?
 - ▶ How to ensure that these officials appropriately distribute taxes without corruption?
- We will cover topics on public service and personnel in developing countries